Let k be a base field. Observe: If A is smooth finite dim algebra/k, then $HH_*(A)$ is finite dim.

Hain's conj: If A is finite dim alg, with finite dim $HH_*(A)$ $\overset{?}{\Rightarrow}$ A is smooth?

Thm (Bochao Kong, L, Yu Shen) the Hain's conjecture is not true.

$$\begin{array}{ccc} Perf(Perf(\Gamma)/[1]) & \xrightarrow{\sim} & Sg(\Lambda) \\ \downarrow & & \downarrow \\ Perf(Perf(\Gamma)/[1])/\langle P_1,\dots,P_{13}\rangle & \xrightarrow{\sim} & Sg(\Lambda)/\langle P_1,\dots,P_{13}\rangle \xrightarrow{\sim} Sg(A) \end{array}$$

$$Perf(Perf(\Gamma)/[1])/\langle P_1,\dots,P_{13}\rangle \overset{SII}{=} Perf((Perf(\Gamma)/\langle P_1,\dots,P_{13}\rangle)/[1])$$

this quotient is designed so that $\simeq P$, $P\neq 0$, $HH_*(P)=0$.

DG categories.

Def: A DG (differential graded) algebra (A,d), here A is a graded algebra, $d: A\to A$ is grading $+1$ map, such that $d^2=0$, $d(xy)=d(x)y+(-1)^{|x|}x\,dy$.

Def (DG categories): A DG category $\mathcal{C}$ is a category enriched in k-complexes. That is: for $\forall X, Y \in \mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(X,Y)$ is a k-complex, and the composition

$$\mathrm{Hom}_{\mathcal{C}}(Y,Z) \otimes \mathrm{Hom}_{\mathcal{C}}(X,Y) \longrightarrow \mathrm{Hom}_{\mathcal{C}}(X,Z)$$

is a morphism of k-complexes ($\Leftrightarrow$ Leibniz rule:

$$d(gf) = (dg)\cdot f + (-1)^{|g|} g\, df.$$

A DG functor $F: \mathcal{A} \to \mathcal{B}$ is a functor s.t.

$$F_{XY}: \mathrm{Hom}_{\mathcal{A}}(X,Y) \longrightarrow \mathrm{Hom}_{\mathcal{B}}(FX, FY)$$

is a degree 0 chain map, compatible with id, composition.

A DG algebra is a DG category with one object.

Ex: Let R be an ordinary algebra. $\mathcal{C}_{dg}(\mathrm{Mod}\, R)$:

Obj = complexes of right R-modules,

$$\mathrm{Hom}(X,Y)^n = \prod_{p \in \mathbb{Z}} \mathrm{Hom}_R(X^p, Y^{p+n})$$

$$d(f) = d_Y \circ f - (-1)^n f \circ d_X$$

Def: Let $\mathcal{C}$ be a DG category. A right DG $\mathcal{C}$-module is a DG functor $M: \mathcal{C}^{op} \to \mathcal{C}_{dg}(k)$

$\mathrm{Mod}_{dg}(\mathcal{C}) = \overset{DG}{=}$ category of right $\mathcal{C}$-modules.

For $\forall X \in \mathcal{C}$, let $h_X = \mathrm{Hom}_{\mathcal{C}}(-, X)$. These are representable modules

$\mathcal{C} \xrightarrow{h_X} Mod_{dg}(\mathcal{C})$ Yoneda embedding, fully faithful.

Def: A DG $\mathcal{C}$-module M is semi-free, if it admits

$0 = F_0 M \subseteq F_1 M \subseteq \cdots \subseteq \bigcup_{i \geq 0} F_i M = M$, s.t.

every factor $F_i M / F_{i-1} M$ is $\bigoplus_{X \in \mathcal{C}} h_X[i]$

It is called finite, if filtration is finite, and factors are all finite direct sum.

A DG $\mathcal{C}$-module is called perfect, if it is quasi-iso to a direct summand of a finite semi-free $\mathcal{C}$-module.

Write $\mathcal{P}erf(\mathcal{C})$ = DG model for all perfect modules

$H^0(\mathcal{P}erf(\mathcal{C})) = Perf(\mathcal{C})$.

Example: $A = Q/I$ be a quiver with relations.

View as DG cat: obj = vertices.

$Hom_A(i,j) = e_j A e_i$, $d = 0$

The representables are $P_i = Hom_A(-, i)$.

Def: A DG category $\mathcal{C}$ is pretriangulated, if it is closed under shifts and cones.

Cone: $f: X \to Y$, $cone(f) = Y \oplus X[1]$,

$$d_{cone(f)} = \begin{pmatrix} d_Y & f \\ 0 & d_{X[1]} \end{pmatrix}.$$

If $\mathcal{C}$ is pretriangulated, then $H^0(\mathcal{C})$ is triangulated.

$H^0(\mathcal{C})$: Obj = Obj $\mathcal{C}$. (homotopy category)

$\mathrm{Hom}_{H^0(\mathcal{C})}(X,Y) = H^0(\mathrm{Hom}_{\mathcal{C}}(X,Y))$

Def: $\mathcal{C}$ is idempotent complete, if in $H^0(\mathcal{C})$, every idempotent splits.

Lemma: $\mathcal{C}$ = small DG category.

$\langle \mathcal{C} \rangle$ (pretriangulated idempotent complete) $= \mathrm{Perf}(\mathcal{C})$

Pf: since representables $h_X \in \mathcal{C}$, so all finite semifree modules $\in \langle \mathcal{C} \rangle$. since it is idempotent complete, direct summands $\in \langle \mathcal{C} \rangle$ so

$\mathrm{Perf}(\mathcal{C}) \subseteq \langle \mathcal{C} \rangle$. $\langle \mathcal{C} \rangle \subseteq \mathrm{Perf}(\mathcal{C})$.

Thm (Keller "deriving DG categories" 1994)

Let $\mathcal{C}$ be a small pretri idempotent complete DG category. $G \in \mathcal{C}$ is a classical generator of $H^0(\mathcal{C})$. Put $A = \mathrm{End}_{\mathcal{C}}(G)$ as a DG algebra. Then the DG functor

$F: \mathcal{C} \xrightarrow{\simeq} \mathrm{Perf}(A)$ is a quasi-equivalence

$X \longmapsto \mathrm{Hom}_{\mathcal{C}}(G, X)$.

Drinfeld quotient

Let $\mathcal{C}$ = DG category, $\mathcal{B} \subseteq \mathcal{C}$ a full subcategory

The Drinfeld quotient is: $\mathcal{C}/\mathcal{B}$

$\mathrm{obj}(\mathcal{C}/\mathcal{B}) = \mathrm{obj}(\mathcal{C})$

Hom complexes:

(1) For every $B \in \mathcal{B}$, add a morphism $\varepsilon_B : B \to B$, s.t. $d(\varepsilon_B) = id_B$.

(2) Take the DG category freely generated by adding theses

Then in $H^0(\mathcal{C}/\mathcal{B})$: $\forall B \in \mathcal{B}$, $B \cong 0$.

For $\forall$ small DG category $\mathcal{C}$, $Ac(\mathcal{C}) \subseteq Mod_{dg}(\mathcal{C})$ be full DG subcat of acyclic modules.

$D(\mathcal{C}) = H^0(Mod_{dg}(\mathcal{C}) / Ac(\mathcal{C}))$ derived category

Def: A DG $\mathcal{C}$-module Y is called h-proj (h-inj) if for $\forall N \in Ac(\mathcal{C})$, we have

$Hom_{Mod_{dg}(\mathcal{C})}(Y, N)$ is acyclic.

(h-inj $Hom_{Mod_{dg}(\mathcal{C})}(N, Y)$ is acyclic).

$\forall X \in Mod_{dg}(\mathcal{C})$ is qiso to a h-projective object PX.

For a DG functor $F: \mathcal{C} \to \mathcal{D}$,

$LF: D(\mathcal{C}) \to D(\mathcal{D}) : X \longmapsto F(PX)$

$RF: \quad X \longmapsto F(IX)$

Def: Let $\mathcal{C}$ be a triangulated category. Say $\mathcal{E}$ is a DG enhancement of $\mathcal{C}$, if $\mathcal{C} \simeq H^0(\mathcal{E})$

$F: \mathcal{C} \to \mathcal{D}$ triangulated functor, we say F lifts to DG enhancements, if there are DG enhancements $\mathcal{E}, \mathcal{D}$ and $\mathcal{F}: \mathcal{E} \to \mathcal{D}$, s.t. $H^0(\mathcal{F}) = F$.

Remark: DG enhancements may not be unique. (Rizzardo - Van den Bergh 2019).